

9- Which one is between pulmonary artery and aorta?

- ☒ A- Ligamentum arteriosum ✓
- B- Oblique pericardial sinus
- C- Transverse pericardial sinus
- D- Ligamentum venosum

A

10- First posterior intercostal artery is a branch of the

- ☒ A- Highest intercostal a. ✓
- B- Thoracic aorta
- C- Thyrocervical trunk
- D- Common carotid a.

A

11- Which muscle is innervated by the long thoracic nerve?

- A- Pectoralis major m.
- B- External intercostal m.
- C- Innermost intercostal m.
- ☒ D- Serratus anterior m. ✓

D

12- In most people SA node and AV node are supplied by

- A- Circumflex a.
- B- Anterior interventricular a.
- ☒ C- Right coronary a. ✓
- D- Left coronary a.

C

Heart Physiology: Dr. Imani

13- When does the aortic valve close?

- A- Left ventricular pressure becomes more than aorta pressure
- ☒ B- Aorta pressure becomes higher than left ventricular pressure ✓
- C- At the rapid ejection phase of the systole
- D- At the slow ejection phase of the systole

B

14- Which ion currents leads to the depolarization phase of the action potential in atrial muscle cells?

- A- Ca^{2+} efflux
- B- Ca^{2+} influx
- C- Na^{+} efflux
- ☒ D- Na^{+} influx ✓

D

15- What is the effect of sympathetic beta receptor stimulation in the heart?

- A- A decrease in heart rate
- ☒ B- An increase in prepotential Steep ✓
- C- Hyperpolarization
- D- An increase in K^{+} efflux

B

16- What is the axis of lead aVR in a normal heart strip?

- ☒ A- 210°
- B- 120°
- C- 60°
- D- -30°

A

17- What is the ST segment on an electrocardiogram?

- A- From the beginning of the QRS to the beginning of the T wave
- B- From the end of the QRS complex to the end of the T wave
- ☒ C- From the end of the QRS complex to the beginning of the T wave
- D- From the beginning of the QRS to the end of the T wave

C

18- In which phase of the systole, the largest amount of blood expels into the aorta?

- A- Isovolumic contraction
- ☒ B- Rapid ejection
- C- Slow ejection
- D- It is the same in all phases.

B

19- What is the preload in the ventricle?

- ☒ A- End diastolic volume
- B- End systolic volume
- C- Aorta and pulmonary arterial pressures
- D- Maximum ventricular contractile force during systol

A

20- Choose the correct sequence of action potential conduction along the conductive tissues of the heart?

- A- SA node $\rightarrow$ AV node $\rightarrow$ Bundle of His $\rightarrow$ Purkinje fibers $\rightarrow$ Bundle branches
- B- SA node $\rightarrow$ AV node $\rightarrow$ Bundle branches $\rightarrow$ Bundle of His $\rightarrow$ Purkinje fibers
- ☒ C- SA node $\rightarrow$ AV node $\rightarrow$ Bundle of His $\rightarrow$ Bundle branches $\rightarrow$ Purkinje fibers
- D- SA node $\rightarrow$ AV node $\rightarrow$ Purkinje fibers $\rightarrow$ Bundle of His $\rightarrow$ Bundle branches

C

21- In cardiac muscle cells, approximately what percentage of cytoplasmic Ca^{2+} required for contraction is released from the sarcoplasmic reticulum and the extracellular fluid, respectively?

- A- 30%-70%
- ☒ B- 70%-30%
- C- 40%-60%
- D- 60%-40%

B

22- What is the function of the chordae tendineae?

- A- Transport blood
- B- Initiate depolarization
- ☒ C- Prevent valve bulging toward the atria during ventricular contraction
- D- Control heart rhythm

C

Anatomy: Dr. Hasanzadeh

- 1- Where is location of the sinoatrial node? A
☒ A- Superior part of the crista terminalis ✓
B- Inferior part of the crista terminalis
C- Medial wall of the right atrium
D- Medial wall of the left atrium
- 2- Where is location of septomarginal (Moderatory) band? C
A- Right atrium
B- Left atrium
☒ C- Right ventricle ✓
D- Left ventricle
- 3- Where is location of the oblique (Marshall) vein? B
A- Posterior surface of the right atrium
☒ B- Posterior surface of the left atrium ✓
C- Inferior border of the heart
D- Between left atrium and left ventricle
- 4- Which nerve has relationship with posterolateral surface of the superior vena cava? D
A- Left vagus n.
B- Right vagus n.
C- Left phrenic n.
☒ D- Right phrenic n. ✓
- 5- Which structure is located in the superior and posterior mediastinum? C
A- Trachea
B- Superior vena cava
☒ C- Esophagus ✓
D- Brachiocephalic trunk
- 6- Where is location of the thoracic duct? B
A- Anterior mediastinum
☒ B- Posterior mediastinum and superior mediastinum ✓
C- Anterior mediastinum and superior mediastinum
D- Middle mediastinum
- 7- Which one is a main characteristic of the thoracic vertebrae? D
A- Long spinous process and foramen transversarium
B- Kidney shape large body
C- Triangular vertebral foramen
☒ D- Articular facet on the lateral surface of body ✓
- 8- Where is location of the apex of the heart? C
A- 5th right intercostal space, 8 to 10 cm from midline
B- 4th right intercostal space, 8 to 10 cm from midline
☒ C- 5th left intercostal space, 8 to 10 cm from midline ✓
D- 4th left intercostal space, 8 to 10 cm from midline

23- In which type of vessels, great reduction in blood pressure is seen?

- A- Vena cava
- ✓ ☒ B- Arterioles
- C- Aorta
- D- Large arteries

B

24- During physical activity in a healthy individual, which of the following is more likely to occur in skeletal muscles during exercise?

- A- Vascular conductance
- B- Blood flow
- C- Arterial diameter
- ✓ ☒ D- All of the above

D

25- In which of the following is the tendency for turbulent flow greater?

- A- Arteries
- B- Capillaries
- C- Small arteries
- ✓ ☒ D- Aorta

D

26- Which component of the circulatory system contains the highest percentage of total blood volume?

- A- Arteries
- B- Capillaries
- ✓ ☒ C- Veins
- D- Pulmonary circulation

C

27- A twofold increase in which of the following will result in the greatest increase in oxygen transfer across the capillary wall?

- A- Capillary hydrostatic pressure
- B- Intercellular clefts in the capillary wall
- ✓ ☒ C- Oxygen concentration gradient
- D- Colloid osmotic pressure of plasma

C

28- Under controlled conditions, vascular blood flow is 100 mL/min with a pressure gradient of 50 mmHg. After tripling the diameter of the vessel, what will be the approximate flow in the vessel?

- A- 3200 mL/min
- ✓ ☒ B- 8100 mL/min
- C- 1600 mL/min
- D- 2500 mL/min

B

29- Which of the following capillaries has the lowest permeability to plasma molecules?

- A- Glomerular
- B- Liver
- C- Muscle
- ✓ ☒ D- Brain

D

30- Which of the following vessels has the largest total cross-sectional area in the circulatory system?

- A- Aorta
- B- Small arteries
- ☒ C- Capillaries ✓
- D- Venules

C

31- Which of the following causes an increase in pulse pressure in an individual?

- A- Decreased stroke volume
- B- Increased arterial compliance
- C- Hemorrhage
- ☒ D- Atherosclerosis ✓

D

32- Which part of the circulatory system has the highest vascular compliance?

- A- Capillaries
- B- Small arteries
- ☒ C- Veins ✓
- D- Aorta

C

33- A decrease in which of the following causes an increase in lymph flow?

- A- Capillary hydrostatic pressure
- ☒ B- Colloid osmotic pressure of plasma ✓
- C- Lymphatic pump activity
- D- Arterial diameter

B

34- Calculate the filtration coefficient based on ml/min/mmHg for a capillary bed with the following data:

Colloid osmotic pressure of plasma = 30 mmHg,

Capillary hydrostatic pressure = 40 mmHg,

Interstitial hydrostatic pressure = -5 mmHg,

Interstitial colloid osmotic pressure = 5 mmHg,

Filtration rate = 200 mL/min,

Venous hydrostatic pressure = 10 mmHg.

A

- ☒ A- 10
- B- 15
- C- 20
- D- 25

Blood: Dr. Ranjbaran

35- What causes decreased venous hydrostatic pressure in the legs?

- A- Increased right atrial pressure
- B- Pregnancy
- ☒ C- Movement of skeletal leg muscles ✓
- D- Presence of ascetic fluid in the abdomen

C

36- What is the main function of red blood cells?

- A- Fighting infections
- B- Producing hormones
- ☒ C- Carrying oxygen
- D- Making antibodies

C

37- Which protein inside red blood cells binds to oxygen?

- A- Albumin
- ☒ B- Hemoglobin
- C- Fibrinogen
- D- Myosin

B

38- What is "margination" in the process of white blood cell response to inflammation?

- A- Movement of white blood cells through capillary walls into tissues
- B- Destruction of pathogens by white blood cells
- ☒ C- Accumulation of white blood cells along the walls of blood vessels
- D- Release of histamine by mast cells

C

39- What is "chemotaxis" in the context of white blood cells?

- A- The process of oxygen binding to hemoglobin
- ☒ B- Movement of white blood cells toward the site of infection due to chemical signals
- C- Formation of blood clots
- D- Breakdown of pathogens by enzymes

B

40- Which of the following is a natural anticoagulant that prevents blood clotting?

- A- Thrombin
- B- Fibrinogen
- ☒ C- Heparin
- D- Calcium

C

41- What is the first step in the process of hemostasis?

- A- Platelet aggregation
- B- Coagulation
- C- Fibrin formation
- ☒ D- Vascular spasm

D

Histology and Embryology: Dr. Asadi Golshan

42- The following embryonic structures are involved in the formation of the foramen ovale EXCEPT:

- ☒ A- ostium primum
- B- ostium secundum
- C- septum primum
- D- septum secundum

A

43- Which of the following is NOT part of Tetralogy of Fallot?

- A- Ventricular septal defect
- B- Pulmonary stenosis
- C- Hypertrophy of the wall of the right ventricle
- ☒ D- Atrial septal defect

D

44- Aortic arch IV forms the on the right side of the embryo and the on the left.

A- Subclavian artery - Pulmonary Artery

B- Arch of aorta - Pulmonary Artery

☒ C- Subclavian artery - Arch of aorta

D- Arch of aorta - Ductus arteriosus

C

45- The SVC and portal veins are derived from which of the following?

A- Vitelline veins - Umbilical veins

B- Anterior cardinal veins - Posterior cardinal veins

☒ C- Anterior cardinal veins - vitelline veins ✓

D- Umbilical veins - vitelline veins

C

46- What is the primary site at which white blood cells adhere to endothelium and leave the circulation

A- Arterioles

B- Lymphatics

☒ C- Postcapillary venules ✓

D- Capillaries

C

47- An artery has lumen and its is the thickest layer.

A- Smaller - adventitia

B- Larger - adventitia

C- Larger - media

☒ D- Smaller - media ✓

D

48- Which one is the main organ involved in filtration of lymph and blood respectively?

☒ A- Lymph node - spleen ✓

B- Spleen - spleen

C- MALT - lymph node

D- Thymus - spleen

A

49- Which region of lymph node contains lymphoid tissue rich in T cells?

A- All parts

☒ B- Paracortex ✓

C- Medulla

D- Cortex

B

50- Hassall corpuscles are unique to the of the

A- White pulp - spleen

B- Medulla - lymph node

C- Cortex - thymus

☒ D- Medulla - thymus ✓

D